

USB LED Night Light

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A typical USB LED bulb shown in Fig. 1 is a 5V, 5W USB-powered solid-state lamp. At the heart of this light bulb is a circular aluminium PCB made with a bunch of 5730 SMD super-bright LEDs. Typical working voltage of a single 5730 SMD LED is in the range of 2.9V to 3.4V, and its current consumption is about 150mA. Fig. 2 shows some details of SMD LEDs used in the USB LED bulb. This gizmo inspired me to design a distinct USB LED night light with battery backup.

Circuit and working

Circuit diagram of the USB LED night light is shown in Fig. 3. It is

built around USB A-type connector (CON1), diode 1N4007 (D1), bilateral switch IC 74HC4066N (IC1) and 5mm piranha white LED (LED1).

This circuit is wired around the popular quad bilateral switch 74HC4066N. Note that, only two of its switches (IC1A and IC1B) are used here. Front-end of the circuit is a standard USB (A) male connector.

During day time, DC voltage available from a standard USB port (or from a 5V solar panel) can be used to charge the rechargeable battery (BATT.1) of the circuit. The battery gets charged through diode D1 connected in series. Since IC1B

is closed by the input voltage, IC1A remains in open state and LED1 rests in off condition.

However, if voltage supplied by the USB port/solar panel is not available, or if it drops to about 1.3V, voltage at the control input of IC1B drops to zero, causing the switch to open. Consequently, IC1A closes, connecting LED1 to the battery through resis-

tor R2, which sets LED1 current to a safe value. Intensity of LED1 may be adjusted by changing the value of R2, observing a maximum current of 20mA through IC1A. Switch S1 prevents the battery from being discharged if the circuit is not in use for some reason.

The 5mm (4-pin) piranha white LED used as the light source, being square in shape, can sit flush against the circuit board, saving a



Fig. 1: Typical USB LED bulb

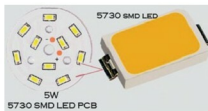


Fig. 2: SMD LED used in a USB LED bulb



Fig. 4: Pin details of the piranha white LED

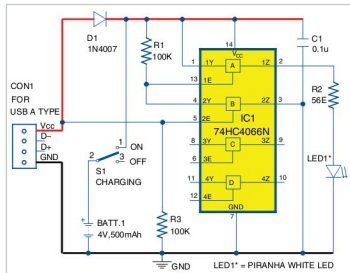


Fig. 3: Circuit diagram of the USB LED night light

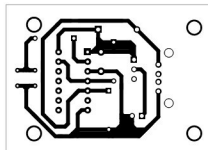


Fig. 5: Actual-size PCB pattern of the USB LED night light

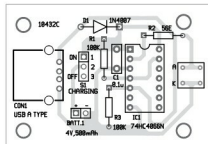


Fig. 6: Component layout of the PCB